Question 1

1.1

S is the initial state where the trading request has been received and no processing checks have been completed yet.

G is the goal state, the trade has passed the required checks and is approved for execution.

The possible actions are the movements from one processing state to another where the agent can move from S to A for market data validation or from B to D for the liquidity check. Each movement follows an edge in the graph and has an associated processing cost.

The path cost is the total cost of all the actions taken along a route, S to B to D has a cost 4 + 1 = 5 so the agent can use these costs to compare different routes and find the route with the lowest total cost.

Abstaction is useful because it removes unnecessary details and focuses only on the information needed to solve the problem whereby techtrade does not need to represent every detail of how employees perform each check, it only needs to represent the processing states, possible movement between them and their costs. This makes the business process easier to represent as a search graph and allows search algorithms to be applied to it.

1.2

step	node	New frontier
1	S(0)	A (2), B(4)
2	A(2)	C (4), D (7)
3	B(4)	D updated to 5 (via B), E (8)
4	C(4)	F (8, via C)
5	D(5)	F updated to 7 (via D), G(12, via D)
6	F(7)	G updated to 9 (via F)
7	E(8)	G via E = 11 not better, discarded
8	G(9)	Goal reached

S – B – D – F – G

Total Cost: 4 + 1 + 2 + 2 = 9

1.3

From S, the choices are A and B of which B has the lower heuristic, so B is chosen.

From B, the choices are D and E which is tie , D is chosen.
From D, the choices are F and G which has the lowest possible heuristic value G is chosen directly and the goal is reached.

S – B – D – G

Total cost: 4 +1 + 7 =12
This route is not optimal because greedy best first search only pays attention to how close a node looks to the goal and never checks how expensive it was to get there. Going straight from D to G looks attractive because h(G) = 0 but the edge D -G = 7 but the algorithm never considers the alternative of going D - F - G which is 2 + 2 = 4 from D and give the true optimal route as S - B - D - F - G = 9.

1.4

A * expands the node with the lowest f(n) = g(n) + h(n) combining the cost already paid with the estimated cost still to go which is why it can find the optimal route while still being guided by the heuristic.

Node	g(n)	h(n)
S	0	6
A	2	5
B	4	4
C	4	4
D	5	2
E	8	2
F	7	1
G	9	0

[S(f(n)] = 0 + 6

= 6

[A(f(n)] =2 + 5

= 7

[B(f(n)] = 4 + 4

= 8

[C(f(n)] = 4 + 4

= 8

[D(f(n)] = 5 + 2

= 7

[E(f(n)] = 8 + 2

= 10

[F(f(n)] = 7 + 1

= 8

[G(f(n)] = 9 + 0

= 9

Total cost = 4 + 1 + 2 + 2 = 9
Final route: S - B - D - F – G = 9

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Question 2
2.1

An agent should be robo-advisor since it is a learning agent as the study states that the firm wants the advisor to improve its recommendations over time based on the outcomes of the previous advice and on client feedback. This is what separates a learning agent from other agents, it does not just act on fixed rules or a single decision rule, it changes its own behaviour as a new experience comes in. Underneath this learning process, it would still need to weigh up different possible recommendations against each clients risk tolerance and expected return which is a utility based style of decision making but the overall architecture the firm is describing, a learning agent is one that adapts based on outcomes and feedback.

However a simple reflex agent would not be a good idea for this system as it chooses action purely from the current percept using fixed condition action rules and it has no way of using the outcome of a previous recommendation to change how it behaves in future. But since the robo advisor must specifically learn from whether past recommendations worked and whether cilents accepted them, a simple reflex agent could never meet this requirement as it would keep repeating the same recommendation pattern even if it kept producing poor outcomes.

Question 2

2.1

A learning agent is suitable because it can improve its performance by learning from previous experiences. The robo-advisor in this study receives information about the client's risk profile, current holdings and market movements. It also considers the results of previous recommendations and client feedback. The system is expected to use this information to improve its future recommendations.

A goal based agent would be less best for this study even though it can select actions that help achieve a specific objective such as increasing investment returns, it does not inherently learn from experience or adapt its decision making process based on past outcomes. The robo-advisor operates in a dynamic financial environment where market conditions and client preferences change over time, therefore the ability to learn from historical performance and client feedback is essential. A learning agent is better suited because it can adapt and improve its recommendations whereas a goal based agent would focus only on achieving predefined goals without continuously enhancing its knowledge and performance.

2.2

This component is responsible for selecting the action that the agent will perform. In the robo advisor, it analyses information such as the client's risk tolerance, investment portfolio and current market conditions before recommending suitable portfolio adjustments.

Continuous improvement is achieved through the learning element which analyses the results of previous recommendations and considering client feedback and can use this experience to improve the investment advice given in the future.

The critic evaluates the results of the agent's decisions and determines how successful they were as it may assess whether a recommendation produced favorable returns or whether the client accepted the suggested portfolio changes.

The problem generator encourages the agent to consider different actions instead of always relying on the same strategies and could involve considering different portfolio allocations that are still suitable for the client's risk profile, this gives the system opportunities to learn which strategies are more effective. These four components enable the robo advisor to learn from experience, adapt to changing market conditions and provide increasingly effective recommendations which requires the advisor to improve its recommendations based on previous outcomes and client feedback.

2.3

a. A rational agent is an agent that selects the action expected to maximize its performance measure based on its percept sequence, prior knowledge and available actions.

b.

The behavior is not rational, the performance measure is to maximize long term returns while remaining within the client's risk tolerance. Since the client is risk averse, recommending a high risk product solely because it recently produced high returns does not support this objective. The agent's percept sequence includes the client's risk profile, its prior knowledge should indicate that recommendations need to match the client's risk tolerance and it has other available actions such as recommending lower risk investments. So the recommendation does not maximize the stated performance measure and is not rational.

c.
A simple reflex agent selects actions based mainly on the current percept, it follows condition action rules and does not maintain information about previous states. A model based reflex agent however maintains an internal state that represents aspects of the environment that cannot be determined from the current percept alone.

The robo advisor needs to maintain an internal state because its decisions depend on more than current market conditions and must consider factors such as the client's risk profile, current holdings, previous recommendations and the outcomes of those recommendations. A simple reflex agent could respond to market changes using fixed rules but this would not be sufficient for the robo advisor. By maintaining an internal state, the system can consider past information and provide more informed recommendations which is essential because the scenario requires the advisor to improve its recommendations using previous outcomes and client feedback.

IBM (2025) What is AI agent learning? IBM Think. Available at: https://www.ibm.com/think/topics/ai-agent-learning [Accessed: 22 August 2026].

Question 3

3.1

A Bayesian network represents a joint probability distribution using a graph and conditional probability tables (CPTs) instead of listing every possible combination of variables in one large table which makes the representation more compact especially when variables are conditionally independent.

Nodes represent random variables in the problem and are economic downturn (E), job loss (J), missed payment (M) and loan default (D).

Directed edges show the dependency between variables, the edge E - J represents the relationship between an economic downturn and job loss. The complete structure is E - J - M - D.

The joint probability can be factorised

$$P(E, J, M, D) = P(E) \times P(J \mid E) \times P(M \mid J) \times P(D \mid M)$$

This factorisation is more compact because each variable is represented only by its conditional relationship with its parent variables than requiring probabilities for every possible combination of variables in the network.

3.2

Yes, D is conditionally independent of E given M.

E-J-M-D

Once M is known, the information about E does not provide any additional information about D through this chain. The direct dependency of D is on M.

$$D \perp E \mid M$$

This means that once the missed-payment status is known, the probability of loan default does not depend directly on whether there was an economic downturn.

3.3

$$P(E, J, M, D) = P(E) \times P(J \mid E) \times P(M \mid J) \times P(D \mid M)$$
$$= 0.20 \times 0.40 \times 0.70 \times 0.60$$

$$= 0.0336$$

3.4

$$P(J) = P(J \mid E)P(E) + P(J \mid \neg E)P(\neg E)$$
$$= (0.40)(0.20) + (0.10)(0.80)$$
$$= 0.16$$
$$P(M) = P(M \mid J)P(J) + P(M \mid \neg J)P(\neg J)$$
$$= (0.70)(0.16) + (0.20)(0.84)$$
$$= 0.28$$

P(J) = 0.16 and P(M) = 0.28

3.5

One strength of a Bayesian network is that it provides a clear and structured way to represent the relationships between different factors that influence credit risk. Here the network shows how an economic downturn can increase the likelihood of job loss which may lead to missed payments and eventually result in loan default. This helps the bank understand how risk factors are connected and supports more informed credit risk assessment and decision making. One limitation is the difficulty of obtaining accurate conditional probability table values as the probabilities in the network must be estimated using historical data or expert knowledge and reliable data may not always be available for every situation. So past data may not accurately reflect future economic conditions or customer behaviour. Although a Bayesian network is a useful tool for modelling and analysing credit risk, its effectiveness depends heavily on the availability of accurate, reliable and representative data for estimating the CPT values.

Question 4

4.1

Every high value transaction is a large transaction

$$\forall x\ (HighValue(x) \rightarrow Large(x))$$

Every transaction that is both large and international must be reported

$$\forall x((Large(x) \wedge International(x)) \rightarrow MustReport(x))$$

Transaction T1 is high value

$$HighValue(T1)$$

Transaction T1 is international

$$International(T1)$$

4.2

$$\forall x\big(HighValue(x) \rightarrow Large(x)\big)$$
$$\forall x\big(\neg HighValue(x) \vee Large(x)\big)$$
$$clause\ 1\ \neg HighValue(x) \vee Large(x)$$

$$\forall x\ \Big(\big(Large(x) \wedge International(x)\big) \rightarrow MustReport(x)\Big)$$
$$\forall x\ \Big(\neg\big(Large(x) \wedge International(x)\big) \vee MustReport(x)\Big)$$
$$\forall x\big(\neg Large(x) \vee \neg International(x) \vee MustReport(x)\big)$$
$$clause\ 2\ \neg Large(x) \vee \neg International(x) \vee MustReport(x)$$

4.3

Unification is the process of finding a substitution that makes two logical expressions identical so that inference can take place which contains variable and constants.

MustReport(x) and MustReport(T1)

MGU = {x/T1} , which indicates that x is replaced with T1 making the 2 literals MustReport(T1)

4.4

Clauses

$$\neg HighValue(x) \vee Large(x)$$
$$\neg Large(x) \vee \neg International(x) \vee MustReport\ (x)$$

Negated Goal:

$$\neg MustReport(T1)$$

Facts:

$$HighValue(T1)$$
$$International(T1)$$

Resolution:

Clause 1 with clause 3

$$Large(T1)$$

Clause 2 with clause 6

$$\neg International(T1) \vee MustReport(T1)$$

Clause7 with clause 4

$$MustReport(T1)$$

Clause 8 with clause 5

Empty clause

4.5

A taxonomy is a hierarchical way of organising knowledge into classes and subclasses. Subclasses inherit the properties of their parent classes, making knowledge representation more structured and efficient. Since a high value transaction is a type of large transaction, any transaction classified as high value automatically inherits the properties associated with large transactions.

Question 5

5.1

Thinking humanly focuses on developing AI systems that imitate the way humans think and reason as it aims to model human thought processes, decision making, learning and problem solving. This approach draws heavily on cognitive science and psychology to understand how people process information and apply that understanding when designing intelligent systems.

The acting rationally approach focuses on creating AI systems that select actions that are most likely to achieve their goals based on the information available to them. The system does not need to think in the same way as a human and should choose the action that is expected to produce the best outcome. Therefore thinking humanly is concerned with replicating human thought processes whereas acting rationally is concerned with selecting the most effective action to achieve a goal.

5.2

The Dartmouth Summer Research Project is widely regarded as the event that established Artificial Intelligence as a formal field of study. It introduced the term Artificial Intelligence and brought together researchers interested in developing machines capable of intelligent behavior.

Deep Blue defeats Garry Kasparov (1997), IBM's Deep Blue became the first computer to defeat the reigning world chess champion, Garry Kasparov in a match under standard tournament conditions. This achievement demonstrated the ability of AI systems to solve complex problems and highlighted the growing capabilities of machine intelligence.

5.3

AI systems can be trained using historical data and if that data contains existing biases, the system may produce unfair credit decisions for certain groups of customers. So if historical lending data reflects past discrimination, an AI system may unintentionally favour some applicants over others which could result in deserving customers being denied credit and may damage the bank's reputation and customer confidence.

A second ethical implication is transparency and accountability where some AI models operate as black boxes making it difficult to explain how a particular credit decision was reached. Customers may lose trust in the bank if they cannot understand why their application was approved or rejected. When the decision making process is unclear, it can be difficult to determine who is responsible when the system makes an incorrect or unfair decision which can reduce customer trust and confidence in the bank's services.

5.4

One important implementation challenge is model monitoring so after an AI system has been introduced into a bank, its performance may change as customer behaviour, economic conditions and the data used by the model change. These changes can cause the model to become less accurate over time and may result in model drift. Therefore the bank does not monitor the model regularly, incorrect predictions could affect important financial decisions and create financial or reputational risks for the bank.

Establishing a continuous model-monitoring process which could include regularly comparing the model's predictions with actual outcomes, checking whether its performance remains within acceptable limits and reviewing the model when significant changes are detected. Regular validation and monitoring would help the bank identify problems early and decide whether the model needs to be adjusted, retrained or replaced.

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